

## SOUTHWEST HAWAII BUOY 51002, USA

WMO: 992430

Lat: 17.043N

Lon: 157.742W

Elev: 0

StdP: 14.70

Time Zone: -11.00 (W11)

Period: 94-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 72.0       | 72.8 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 31.8                  | 75.0 | 30.4 | 74.8 | 18.6                  | 80   | 0.743 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 9             | 2.0                    | 82.7            | N/A  | 82.0 | N/A  | 81.4 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 14.9                 | 90   |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 29.5          | 26.1 | 24.7 | DB                         | 69.2 | 81.4               | 1.7 | 2.0                                                | 67.9 | 82.8       | 67.0 | 84.0       | 66.0 | 85.2       | 64.8 | 86.6 | (4) |
| (5)               |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 77.0   | 75.5  | 75.0  | 75.3  | 75.6  | 76.2  | 77.2  | 78.0  | 78.7  | 79.1  | 78.8  | 77.9  | 76.4 |
|                                                                                  | DBStd       | 1.91   | 1.23  | 1.28  | 1.21  | 1.27  | 1.40  | 1.36  | 1.25  | 1.33  | 1.36  | 1.23  | 1.21  | 1.18 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD50       | 9855   | 790   | 701   | 784   | 768   | 814   | 817   | 869   | 891   | 872   | 893   | 837   | 819  |
|                                                                                  | CDD65       | 4380   | 325   | 281   | 319   | 318   | 349   | 367   | 404   | 426   | 422   | 428   | 387   | 354  |
|                                                                                  | CDH74       | 29049  | 1345  | 976   | 1223  | 1382  | 1896  | 2551  | 3216  | 3728  | 3821  | 3798  | 3038  | 2075 |
|                                                                                  | CDH80       | 589    | 0     | 0     | 0     | 0     | 5     | 6     | 48    | 135   | 216   | 140   | 38    | 1    |
| Wind                                                                             | WSAvg       | 17.9   | 16.6  | 17.3  | 18.1  | 18.7  | 17.7  | 18.0  | 18.2  | 17.9  | 16.6  | 17.7  | 18.8  | 19.1 |
| Precipitation                                                                    | PrecAvg     | 18.1   | 1.8   | 1.8   | 1.9   | 0.7   | 0.6   | 0.4   | 0.9   | 1.7   | 1.8   | 2.1   | 2.1   | 2.2  |
|                                                                                  | PrecMax     | 35.4   | 5.6   | 6.7   | 8.0   | 3.1   | 3.5   | 2.2   | 2.5   | 5.9   | 8.5   | 7.4   | 5.1   | 7.6  |
|                                                                                  | PrecMin     | 8.1    | 0.0   | 0.0   | 0.1   | 0.2   | 0.0   | 0.0   | 0.0   | 0.1   | 0.2   | 0.4   | 0.6   | 0.2  |
|                                                                                  | PrecStd     | 6.2    | 1.6   | 1.7   | 2.0   | 0.8   | 0.7   | 0.4   | 0.7   | 1.4   | 1.7   | 1.5   | 1.2   | 1.9  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 79.3  | 78.5  | 78.5  | 79.2  | 80.6  | 80.7  | 82.0  | 83.0  | 84.2  | 82.5  | 82.1  | 80.3 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | DB     | 78.4  | 78.2  | 78.2  | 78.4  | 80.1  | 80.3  | 81.2  | 82.3  | 83.3  | 82.1  | 81.1  | 79.4 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | DB     | 78.0  | 77.8  | 77.6  | 78.2  | 79.3  | 80.0  | 80.5  | 81.6  | 82.3  | 81.4  | 80.4  | 79.0 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | DB     | 77.4  | 77.3  | 77.3  | 77.7  | 78.4  | 79.5  | 80.2  | 80.7  | 81.2  | 80.9  | 80.0  | 78.4 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 2.3   | 2.4   | 2.2   | 2.1   | 2.2   | 2.2   | 2.1   | 2.0   | 2.0   | 2.2   | 2.3   | 2.5  |
|                                                                                  |             | MCDBR  | 2.4   | 2.7   | 2.3   | 2.3   | 2.8   | 2.4   | 2.3   | 2.4   | 2.7   | 2.8   | 3.0   | 3.2  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.359  | 0.369 | 0.377 | 0.406 | 0.400 | 0.390 | 0.392 | 0.393 | 0.384 | 0.378 | 0.375 | 0.367 |      |
|                                                                                  | taud        | 2.605  | 2.580 | 2.558 | 2.446 | 2.484 | 2.504 | 2.518 | 2.525 | 2.547 | 2.601 | 2.598 | 2.603 |      |
|                                                                                  | Ebn at Noon | 295    | 297   | 297   | 286   | 284   | 284   | 284   | 286   | 290   | 291   | 288   | 289   |      |
|                                                                                  | Edh at Noon | 30     | 31    | 33    | 37    | 35    | 34    | 34    | 34    | 33    | 30    | 30    | 29    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1581   | 1730  | 1924  | 2115  | 2220  | 2236  | 2238  | 2184  | 2071  | 1805  | 1553  | 1444  |      |
|                                                                                  | RadStd      | 78     | 96    | 170   | 128   | 104   | 51    | 69    | 83    | 71    | 65    | 84    | 89    |      |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page